Jonathan D. Shaw

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: https://jonathan-d-shaw.github.io

Education

Ph.D. in Applied Mathematics — *Expected May 2028*
University of Colorado Boulder — Boulder, CO

M.S. in Applied Mathematics — *May 2026*
University of Colorado Boulder — Boulder, CO

B.S. in Applied Mathematics — *May 2023*
University of Colorado Boulder — Boulder, CO
Magna cum laude
Concentration in Aerospace Engineering

Research Interests

- Numerical partial differential equations
- Integral equations
- "Fast" algorithm design
- Multiscale modeling/simulation
- Scattering and inverse problems
- Applications of functional, asymptotic, and numerical analysis

Publications

1. Trujillo, M.; Sajadi, A.; **Shaw, J. D.**; Hodge, B.-M., *Computationally Efficient Analytical Models of Frequency and Voltage in Low-Inertia Systems*, IEEE Transactions on Power Delivery 2026 (: arXiv)
2. Trujillo, M.; Sajadi, A.; **Shaw, J. D.**; Hodge, B.-M., *Analytical Models of Frequency and Voltage in Large-Scale All-Inverter Power Systems*, IEEE Transactions on Power Systems 2025 (: arXiv)
3. **Shaw, J. D.**; Restrepo, J. G.; Rodríguez, N., *A dynamical system model of gentrification: Exploring a simple rent control strategy*, Phys. Rev. E, 2024 (: arXiv)

Awards

- Student Travel Award, Dynamics Days US 2025

Talks & Presentations

Talks

- *A quasi-direct solver for boundary integral equations*, Research Training Group in Numerical and Scientific Computing, Rice University, Houston, TX, October 2026 (contributed talk; travel funding provided)
- *Improving the density interpolation method*, SIAM Graduate Student Chapter Seminar, University of Colorado Boulder, January 2026 (contributed talk)

- *A dynamical system model of gentrification*, Complex/Dynamical Systems Seminar, University of Colorado Boulder, March 2025 (invited talk)
- *A dynamical system model of gentrification*, Dynamics Days US 2025, Denver, CO, January 2025 (contributed talk)
- *On the dynamics of gentrification*, Mathematical Biology Seminar, University of Colorado Boulder, October 2023 (contributed talk)

Poster Presentations

- *A dynamical system model of gentrification*, Dynamics Days US 2023, Denver, CO, January 2023

Research Experience

Department of Applied Mathematics — University of Colorado Boulder
Research Assistant — *May 2022–Present*

QUASI-DIRECT SOLVERS FOR BOUNDARY INTEGRAL EQUATIONS — *May 2026–Present*
Project advisor: *Adrianna Gillman*

IMPROVING THE DENSITY INTERPOLATION METHOD — *August 2025–May 2026*
Project advisor: *Eduardo Corona*

FAST SOLVERS FOR POWER SYSTEMS APPLICATIONS — *May 2023–August 2025*
Project advisor: *Eduardo Corona*
Project collaborators: *Fiona Majeau*, *Bri-Mathias Hodge*, *Amir Sajadi*

GENTRIFICATION DYNAMICS — *May 2022–September 2024*
Project advisors: *Juan Restrepo* and *Nancy Rodríguez*

Los Alamos National Laboratory — Los Alamos, NM
Intern, Physics XTD-PRI — *May 2018–August 2018*

ALGEBRAIC TOPOLOGY FOR COMPUTATIONAL FLUID DYNAMICS
Project advisor: *Roy S. Baty*

Software

Contributions

- chunkIE: A MATLAB integral equation toolbox

Leadership & Service

Society for Industrial and Applied Mathematics (SIAM) — University of Colorado Boulder

President, Graduate Student Chapter — *August 2025–Present*

Industry Liaison, Graduate Student Chapter — *August 2024–May 2025*

Buckskin Youth Leadership Training — Richmond, VA
Senior Staff Member — *January 2016–August 2023*